

CIV102 Project Team 509 Engineering Assembly

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Engineering Drawings

Bridge Final Construction

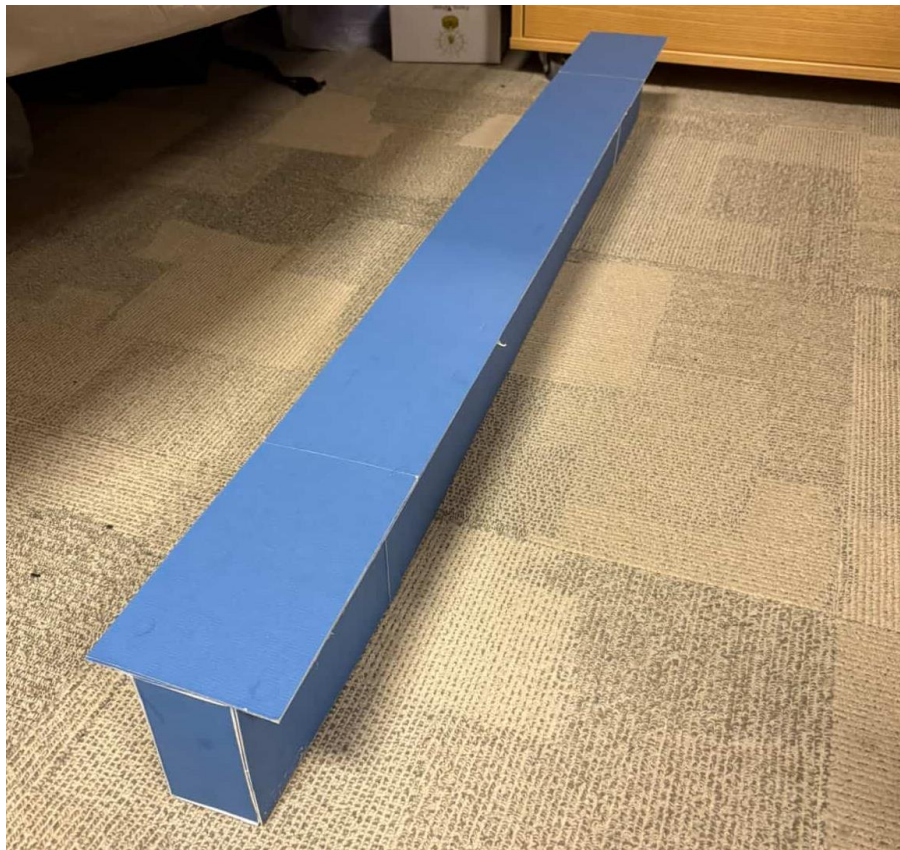


Figure 1: Final Bridge Design after Construction Process.

Bridge Cutout

CutList Optimizer

Stock sheet	1016×810
Used area	802025.6 97%
Wasted area	20934.4 3%
Cuts	55
Cut length	12925
Panels	39
Wasted panels	16

Panel	Qty
200×120	2
850×120	1
300×120	2
650×120	1
355 2×48	1
90×60	2
200×125.8	4
850×125.8	2
70×122.5	10
131.5×70	6
20×116.2	4
20×117.5	4

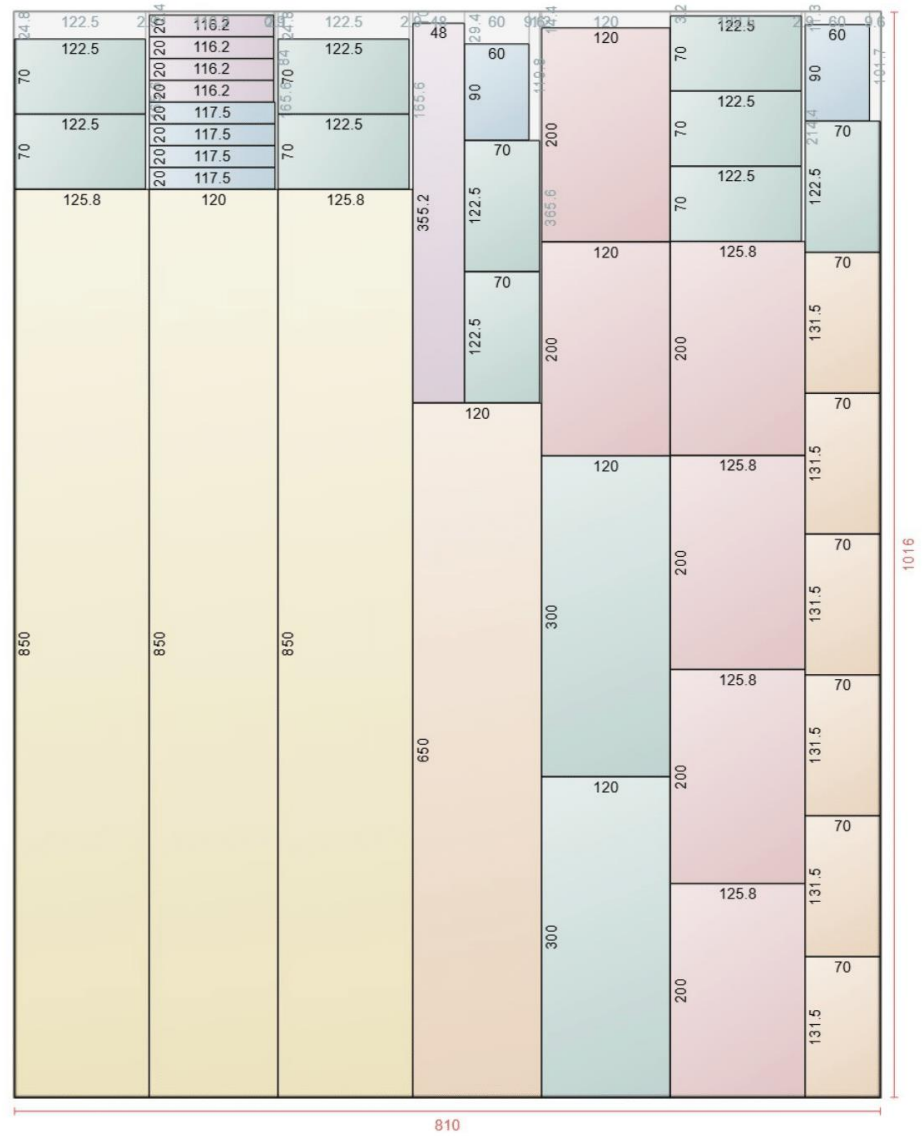


Figure 2: Cutout for bridge generated with cut list optimizer.

Bridge General Layout

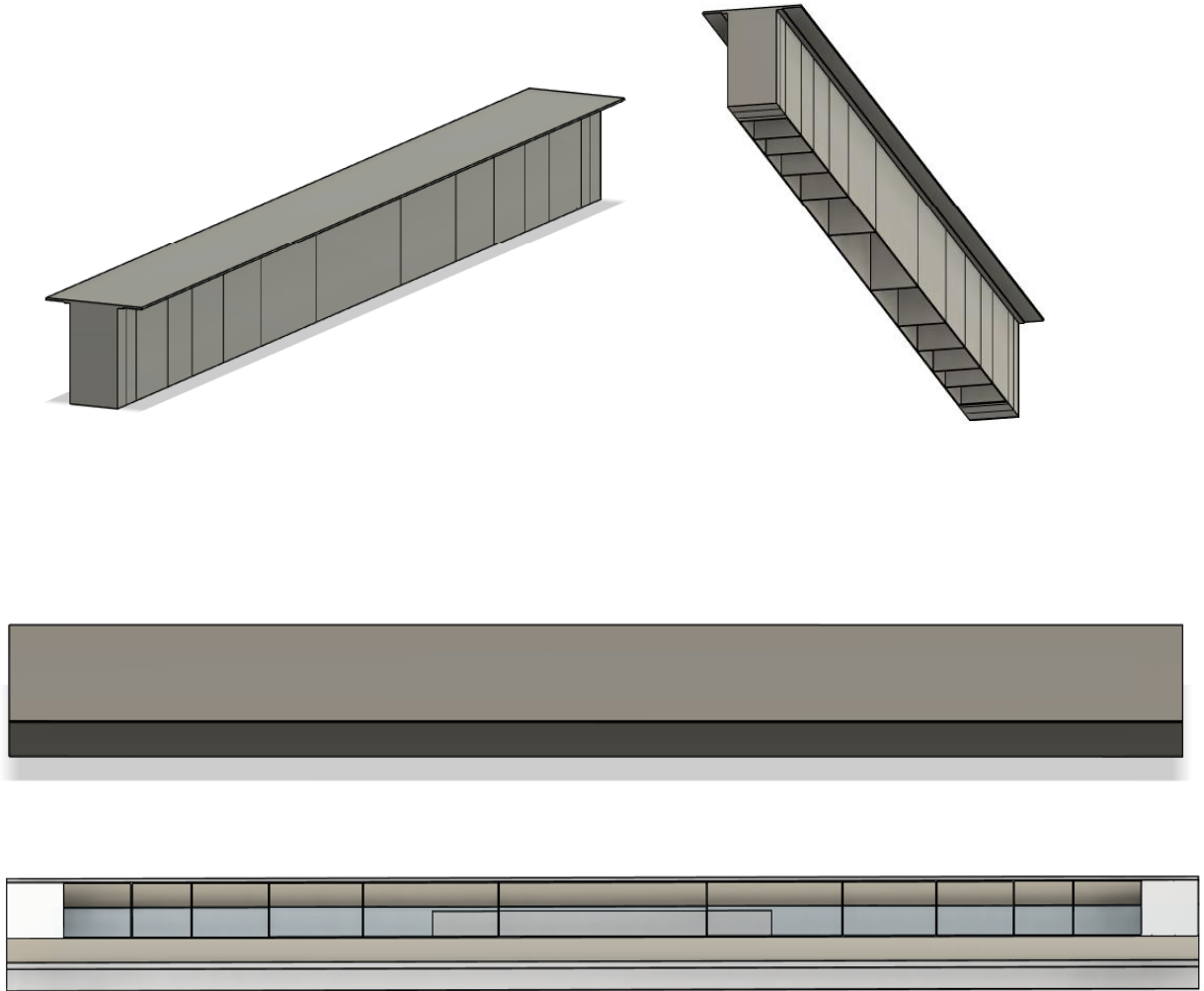


Figure 3: 3D Model of planned bridge, showcasing roof, diaphragms, supports, and cross section in 3D view.

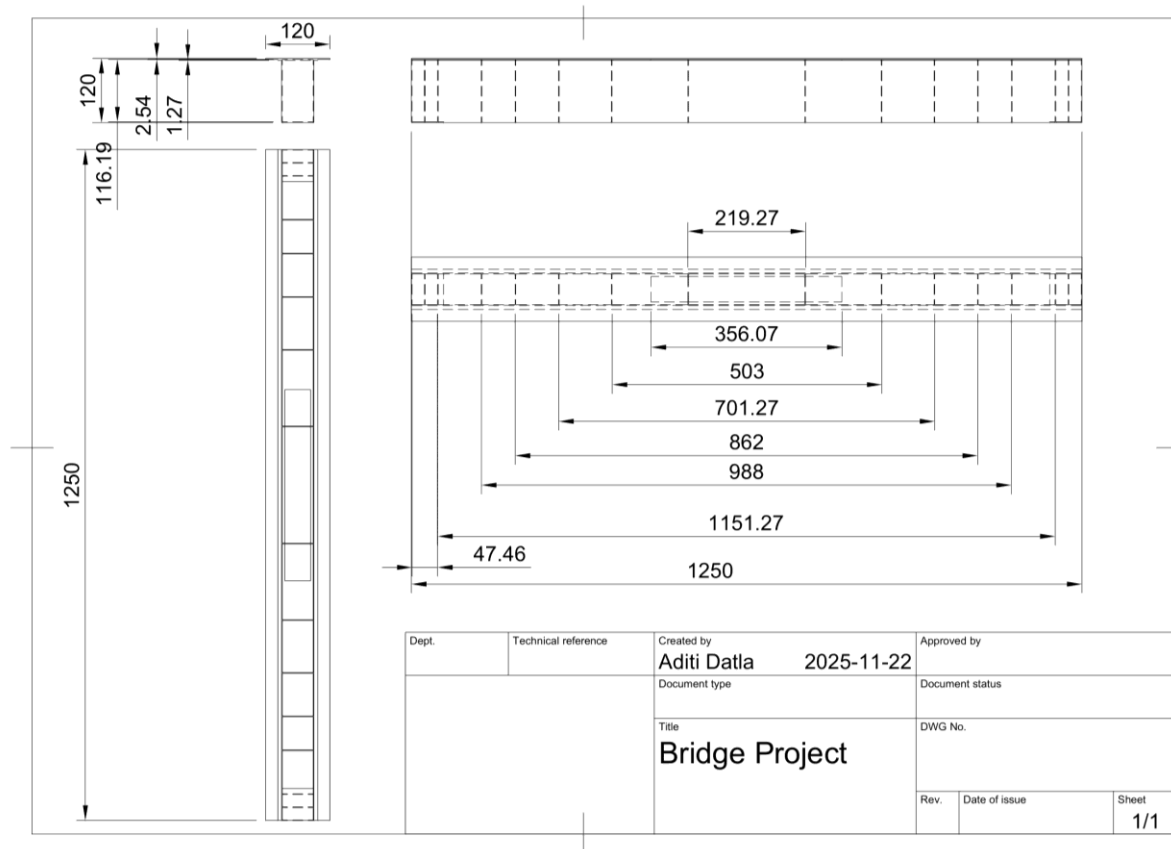


Figure 4: Layout of full bridge in mm dimensions, showcasing front, side, back, and top view.

Bridge Cross Sections

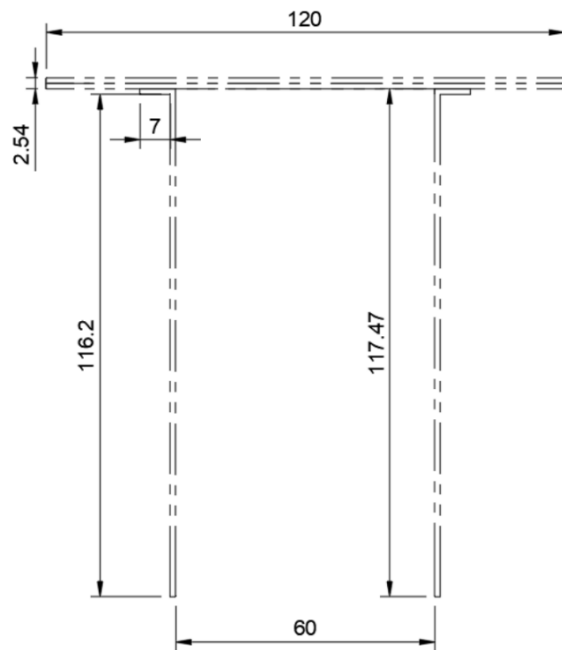


Figure 5: Dimensions of Bridge Cross Section without spine in mm.

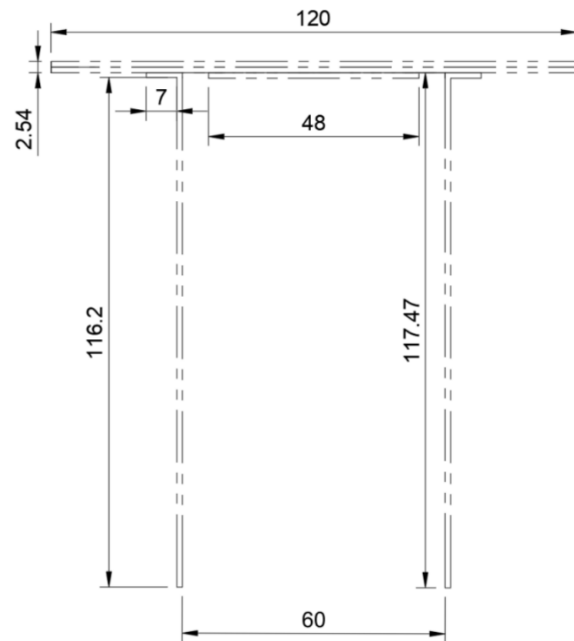


Figure 6: Dimensions of Bridge Cross Section in mm with spine spanning 355mm across center of bridge

Bridge Diaphragms

The diaphragms were spaced quadratically, with closer spacing near the supports and wider spacing toward the center, because this arrangement provides greater stiffness where shear forces are highest while reducing unnecessary material where the demand is lower.

Diaphragm # Position (mm)

1	0
2	25.0
3	50.0
4	131.6
5	195.0
6	277
7	376
8	520
9	739
10	884
11	983
12	1065
13	1128
14	1210
15	1235
16	1260

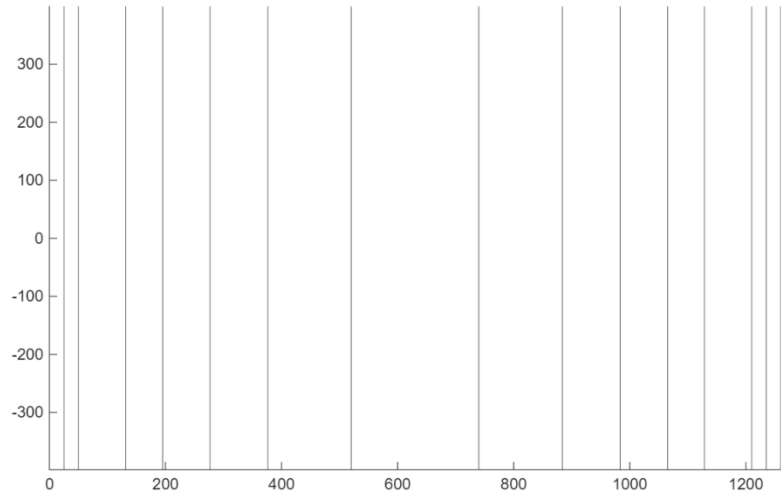


Figure 8: MATLAB code output showing the spacing of the diaphragms across the bridge.

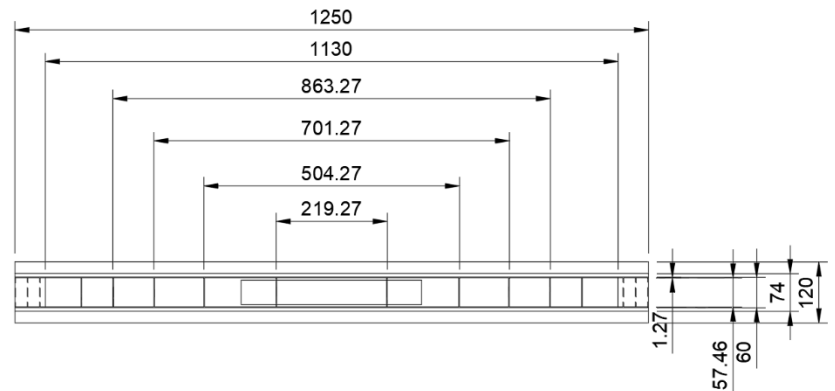


Figure 7: Dimensions and spacing of symmetrical diaphragms across length of bridge.

Bridge Splicing

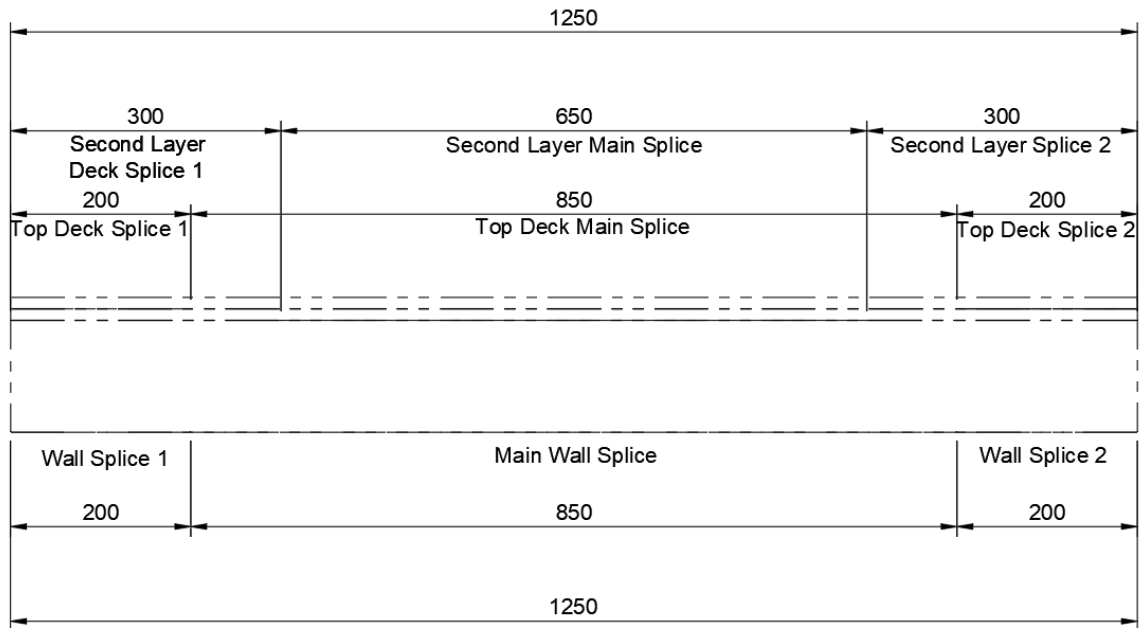


Figure 9: Diagram showing the spacing of splices made for the main walls, top deck, and second layer deck.

Building Process

Overview

The building process of the bridge was all done in our teammate's house with proper care taken for each step. While drawing and cutting was done in this teammate's basement, gluing for the bridge was done in this teammate's garage with the door open to allow proper ventilation for our group's safety.

Drawing

Our team began by gridding our matboard with parallel lines to ensure that our outlines for the pieces of our bridge were precise and rectangular.

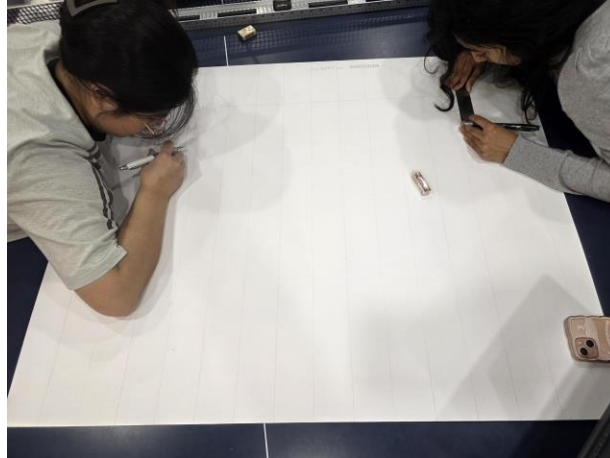


Figure 10: Drawing grid lines on MATBOARD to ensure straight outlines are made (Taken 11/20/2025 - 9:16 PM)

After this, we drew our outlines onto the matboard using a level to ensure that the lines were exact. However, we quickly noticed that our level was slightly damaged and had a curve to it that was affecting our lines. Because of this, we switched to drawing with rulers and followed the grid to complete our outline. Every line was measured twice and corrected to ensure that our pieces were exact.



Figure 11: Outlining shapes onto matboard to be cut out later on. (Taken 11/20/2025 - 10:20 PM)

Cutting

After verifying that all our outlines were accurate, we began to cut out each piece from the matboard using a utility knife. All pieces were cut with a scrap piece of board underneath to avoid damage to the table used.

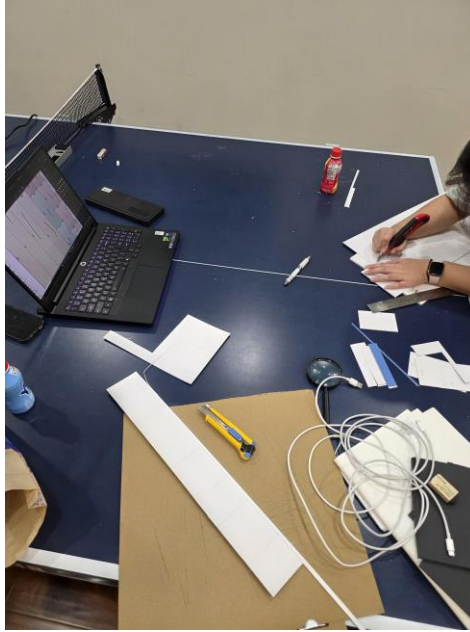


Figure 12: MATBOARD cutting process, following our existing outline. (Taken 11/21/2025 - 3:00 AM)

Gluing

As we prepared to glue our pieces together, we marked out on each piece where the glue will be put exactly and the scoring of all the glue tabs to ensure that no glue wasted since we only had 2 bottles that had to be used fully for the entire bridge. This phase ensured that we would not be stuck without glue later in the process.

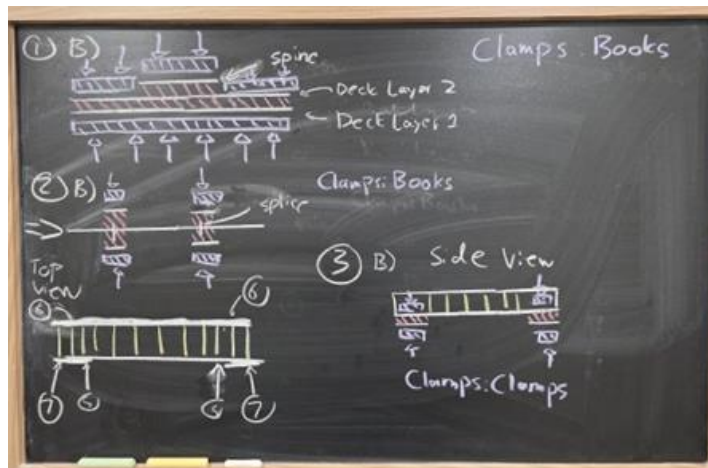
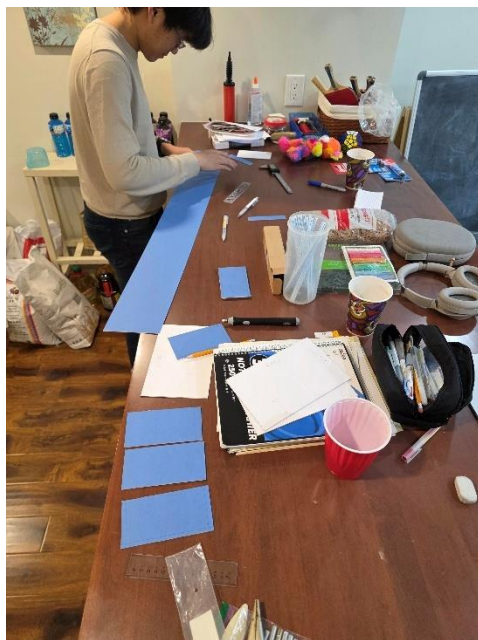


Figure 13-14: Pictures showing the planning process for gluing including the layout on the right and the labelling on the pieces on the left. (Taken 11/21/2025 - 3:00 AM)

Before any gluing began, a tarp was laid over the floor of the garage to protect our workspace from the glue. We began by gluing the diaphragms into our top plate with a thin brush to limit our glue usage.



Figure 15: Our initial workspace setup for gluing with a tarp on the floor to prevent staining in an open garage environment. (11/21/2025 - 3:41 AM)

It was at this stage that we realized that our layout was missing the glue tabs needed to connect the walls to our top plate. Because of this oversight, we used all our remaining scraps of matboards as our new glue tabs to cover the necessary area. As these scraps were around the same length as our glue tabs, we knew it would not impact the glue shear failure significantly.

After we were done gluing our two top plate pieces together, we let them dry on the side with textbooks and weights put on top to compress the two pieces and ensure the glue dries properly



Figure 16: Weights laid across the top deck glued to second layer to apply pressure. (Taken 11/21/2025 - 4:41 AM)

After this, the diaphragms were carefully attached to the walls by their glue tabs.



Figure 17-18: Gluing process of diaphragms onto the walls of the bridge with brush for precision. (Taken 11/21/2025 - 10:09 AM)

When the diaphragms were completed, we left this piece to dry with more weights putting pressure on both sides of the walls against the glue tabs to ensure they dry in a straight format.



Figure 19: Weights applied to both sides of the walls to increase pressure on sides and secure glue. (Taken 11/21/2025 - 11:49 AM)

When both pieces were sufficiently dried, we finally combined the two items by placing the top plate onto the bridge body. Weights were placed across the bridge to apply pressure, along with clamps to keep the top and body together. This configuration was then set to dry over the course of 4 days before the final testing.

While the pieces were drying, we noticed that some glue tabs kept coming loose due to the minimal glue use required in the project. Because of this, we reheated and secured the piece once again with a clamp.



Figure 20: Top deck glued on top of base with weights and clamps securing the position to be glued. (11/21/2025 - 2:22 PM)

When all the construction work was done, both areas utilized were returned to their original state.



Figure 21: Both areas used shown to be free from damage after the construction process. (Taken 11/22/2025 11:09 AM)

Time Log

Tasks	Date	Perry	Sam	Aditi	Hasnain	Total
Deliverable 1						
	11/6/2025	2	2	2	2	8
Planning						
	11/12/2025	1.5	1.5	1.5	1.5	6
	11/17/2025	1	1	1	1	4
	11/19/2025	8	6	6	7	27
	11/21/2025	4	2	2	2	10
Report						
	11/20/2025		6	2	1	9
	11/21/2025		3			3
	11/22/2025		3	6		9
	11/23/2025	2	6	8	2	17
Drawing						
	11/17/2025				2	2
	11/19/2025		1		2	3
	11/20/2025		4	2		6
	11/23/2025		2		1.5	3.5

						0
Cutting	11/19/2025		5			5
Calculations						
Hand Calculations						
	11/21/2025				2	2
	11/19/2025	4			3	7
	11/18/2025				1.5	1.5
	11/23/2025					
Code						
	11/12/2025			9		9
	11/18/2025	10				10
	11/23/2025			2		2
Construction						
	11/19/2025-11/20/2025	16	4	2	15	37
Total		 48.5	 46.5	 43.5	 43.5	182